

**THE IMPACT OF IMMERSIVE TECHNOLOGIES ON STEM
EDUCATION IN SRI LANKA'S**

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Final Dissertation

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DECLARATION

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ABSTRACT

This research examines the effectiveness of Augmented Reality (AR)-based mobile learning in improving Grade 8 students' understanding of astronomy concepts in Sri Lanka. Traditional teaching methods rely on static textbook diagrams and verbal explanations, which often make it difficult for students to visualize the three-dimensional and dynamic nature of the Solar System, leading to reduced engagement and conceptual understanding.

To address this issue, a marker-based AR mobile application was developed using Unity 3D and Vuforia. The application allows students to scan textbook images and interact with 3D models of the Solar System. It includes interactive features such as rotation, zooming, and gamified learning activities including Planet Size Comparison, Planet Order Arrangement, and Fact Matching to enhance engagement and understanding.

A paired sample t-test was used to evaluate the effectiveness of the system, where the same group of students was assessed under two conditions: traditional textbook-based learning and AR-based mobile application learning. Pre-test and post-test scores were collected and compared to measure learning improvement.

The application is designed to be offline-capable and optimized for low-end Android devices, ensuring accessibility in resource-constrained school environments. It is also aligned with the Grade 8 science curriculum for easy classroom integration.

The findings are expected to show that AR-based learning significantly improves student engagement and conceptual understanding compared to traditional methods, demonstrating its potential as an effective educational tool in STEM education.

Keywords: Augmented Reality, Solar System, STEM Education, Mobile Learning, Gamification, Paired t-test

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AR	Augmented Reality
STEM	Science, Technology, Engineering and Mathematics
DDE	Design–Development–Evaluation
UI	User Interface
3D	Three-Dimensional
CPU	Central Processing Unit
RAM	Random Access Memory
APK	Android Package Kit
SLIIT	Sri Lanka Institute of Information Technology
GPS	Global Positioning System
UI/UX	User Interface / User Experience
CSV	Comma Separated Values
t-test	Student’s t-test (paired sample statistical test)
Vuforia	Vuforia Engine (AR development platform)

CHAPTER 1: INTRODUCTION

1.1 Background and Literature Survey

Astronomy is a fundamental component of the secondary school STEM curriculum, particularly at the Grade 8 level, where students are introduced to essential concepts related to the structure, scale, and dynamics of the Solar System. Despite its importance, astronomy remains one of the most challenging topics for students due to its abstract and spatial nature. In Sri Lanka, teaching methods predominantly rely on static textbook diagrams, verbal explanations, and memorization-based learning approaches. While these methods align with curriculum requirements, they often fail to effectively represent the three-dimensional and dynamic characteristics of celestial phenomena, resulting in limited conceptual understanding and reduced student engagement.

Research in science education highlights that students often struggle to construct accurate mental models when learning abstract scientific concepts through two-dimensional representations alone. This limitation is particularly evident in STEM disciplines, where spatial reasoning and visualization play a crucial role in understanding complex systems.

Augmented Reality (AR) has emerged as a powerful immersive technology capable of addressing these challenges by blending virtual content with the real-world environment. AR enables learners to interact with 3D digital objects in real time, thereby improving visualization and engagement. Azuma defines Augmented Reality as a system that combines real and virtual environments, allowing users to interact with both simultaneously in real time [1]. This capability makes AR highly suitable for educational applications where abstract visualization is required.

A growing body of literature supports the effectiveness of AR in education. Systematic reviews indicate that AR-based learning environments significantly improve student motivation, engagement, and conceptual understanding compared to traditional instructional methods [2]. Furthermore, AR has been shown to enhance spatial ability and knowledge retention in science education by providing interactive and experiential

learning opportunities [3]. In STEM education specifically, AR has demonstrated strong potential in improving students' understanding of complex and invisible phenomena, such as planetary motion and astronomical structures [4].

Despite these advantages, existing AR educational systems present several limitations. Many applications are not aligned with national curricula, require high-end hardware, or depend on continuous internet connectivity, making them unsuitable for resource-constrained environments. Additionally, most AR solutions are developed in global contexts and do not adequately address local educational challenges or classroom constraints in developing countries such as Sri Lanka [5].

To address these gaps, this research focuses on the design and development of a marker-based Augmented Reality mobile application tailored specifically for Grade 8 astronomy education in Sri Lanka. The system is designed to be offline-capable, optimized for low-end Android devices, and fully aligned with the national science curriculum. It integrates interactive 3D visualization with gamified learning activities to enhance student engagement, conceptual understanding, and problem-solving abilities.

Furthermore, this study evaluates the effectiveness of AR-based learning by comparing it with traditional teaching methods using a paired sample t-test approach. The same group of students is assessed under both instructional conditions, enabling a direct comparison of learning outcomes and providing measurable evidence of the impact of immersive technologies in STEM education.

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1.2 Research Gap

- **Lack of curriculum-aligned AR learning applications for Sri Lankan Grade 8 STEM syllabus**

Studies show that most AR applications in STEM education are developed as general learning tools and are not directly aligned with specific national curricula, limiting

classroom integration effectiveness [2], [8]. Systematic reviews confirm that AR systems are often designed for broad STEM concepts rather than structured syllabus-based instruction.

- **Limited support for low-resource environments (high device + internet requirements)**

Research indicates that many AR learning systems require high-performance mobile devices, stable internet, or advanced hardware, making them unsuitable for resource-constrained classrooms [6], [7]. Studies highlight that technical requirements remain a major barrier to AR adoption in developing educational contexts.

- **Absence of offline-capable AR educational applications**

Existing AR implementations are predominantly internet-dependent or cloud-supported, which limits usability in rural or under-resourced schools. Literature identifies connectivity and infrastructure dependence as recurring constraints in AR-based STEM learning environments [6], [7].

- **Lack of Sri Lankan-based research on AR in STEM education**

Systematic reviews confirm that most AR-STEM studies are conducted in international contexts (Europe, USA, East Asia), with very limited empirical research in South Asian or Sri Lankan education systems [2], [7]. This creates a contextual research gap in understanding AR effectiveness in local classrooms.

- **Lack of paired sample t-test comparisons between traditional and AR-based learning**

Most AR education studies rely on pre-test/post-test group comparisons or descriptive statistics, with fewer studies applying paired-sample statistical methods to the same

learner group for direct comparison [2], [16]. This limits strong causal evaluation of AR effectiveness over traditional methods.

- **Insufficient integration of gamification + AR visualization + assessment in one system**

Although AR improves engagement and visualization, studies show that most systems focus on either visualization or gamification separately, rather than combining interactive AR + game-based learning + embedded assessment tools [2], [16]. This creates a gap in holistic learning system design.

1.3 Research Problem

Based on the identified gap, the central research problem is formulated as follows:

‘How can a low-cost, offline, curriculum-aligned AR mobile application be designed and developed for Grade 8 astronomy to improve students’ conceptual understanding, engagement, and problem-solving skills while supporting teachers in classroom integration?’

1.4 Research Objectives

Main Objective

The main objective of this study is to design and develop a marker-based Augmented Reality (AR) learning module that enhances the understanding of astronomy-related concepts within the Grade 8 Geography curriculum in Sri Lanka by providing interactive 3D visualization and an engaging learning experience.

Sub Objectives

- To design and develop a marker-based AR system using image targets to visualize key astronomy-related concepts such as the Solar System, the Sun, inner planets, and outer planets.

- To implement interactive 3D models that allow users to explore astronomical objects through functionalities such as rotation, scaling, and zooming.
- To integrate gamified learning activities, including Planet Size Comparison, Planet Order sequencing, and Fact Matching, to enhance student engagement and knowledge retention.
- To develop and incorporate performance-based evaluation mechanisms through quizzes and game interactions to assess student understanding.
- To evaluate the effectiveness of the developed AR module in improving student engagement and conceptual understanding within the Grade 8 Geography curriculum.

CHAPTER 2: METHODOLOGY

2.1 Methodology

Research Design

This study adopts a **Design–Development–Evaluation (DDE)** research approach, which is widely used in educational technology research for developing and evaluating digital learning systems in real classroom environments. The DDE model emphasizes iterative system design, structured development, and empirical evaluation of learning effectiveness [3].

The research is organized into three main phases:

- System Design
- System Development
- System Evaluation

This structured approach ensures that the AR learning system is not only technically functional but also pedagogically effective.

System Development Approach

The proposed Augmented Reality (AR) learning system is developed using **Unity 3D** as the primary development platform and **Vuforia Engine** for marker-based image recognition and tracking. Unity is widely adopted in AR educational applications due to its ability to render real-time interactive 3D environments efficiently across mobile devices [4].

The system implements **marker-based AR technology**, where printed textbook images act as triggers for displaying interactive 3D Solar System models. According to Azuma, AR systems combine real and virtual elements in real time to create immersive learning experiences [1].

Development Process Includes:

- Creation of 3D models of the Solar System (Sun, planets, and related celestial objects)
- Implementation of marker-based AR detection using Vuforia
- Integration of interactive gamified learning activities
- Optimization for low-end Android devices (2–3 GB RAM compatibility)
- Implementation of offline functionality for classroom accessibility

The system ensures that students can interact with astronomical models without requiring high-end hardware or continuous internet connectivity.

System Components

The system is structured into three main components:

1. Mobile Application (User Interface)

- Provides student access to AR lessons and gamified activities
- Displays interactive 3D Solar System models
- Includes intuitive UI designed for Grade 8 learners

- Supports offline usage

2. AR Content Engine

- Built using Unity 3D + Vuforia
- Handles marker recognition and 3D rendering
- Enables real-time interaction with planetary models
- Supports rotation, zooming, and exploration of celestial objects

3. Learning Modules

The system includes structured educational content:

- Solar System & Sun
- Inner Planets
- Outer Planets

Gamified Learning Activities:

- Planet Size Comparison
- Planet Order Arrangement
- Fact Matching

These activities enhance engagement and reinforce conceptual understanding through active learning.

Research Method and Data Collection

A **quantitative research approach** is used to evaluate the effectiveness of the AR-based learning system. The same group of Grade 8 students is assessed under two learning conditions:

- Traditional textbook-based instruction
- AR-based mobile application learning

Student performance is measured using **pre-tests and post-tests**, and learning improvement is analyzed using a **paired sample t-test**, which is suitable for comparing two related samples from the same participants [5].

Data Collected Includes:

- Pre-test scores (traditional learning)
- Post-test scores (AR-based learning)
- Quiz performance within application
- Interaction metrics (completion rate, engagement frequency)

Evaluation Method

The system is evaluated based on:

- Improvement in student academic performance
- Engagement during AR learning activities
- Comparative analysis between traditional and AR-based learning outcomes

A **paired sample t-test** is applied to determine whether there is a statistically significant difference between the two learning methods. This statistical approach is widely used in educational research to measure learning gains from instructional interventions [5].

System Architecture Overview

The system is designed using a layered architecture:

- **Presentation Layer:** Mobile UI and AR visualization interface
- **Application Layer:** Lesson logic, gamification system, and interaction control
- **AR Engine Layer:** Unity + Vuforia tracking and real-time rendering system
- **Data Layer:** Local storage for quiz results and interaction data

This modular structure ensures:

- Efficient performance on low-end Android devices
- Scalability for future expansion
- Smooth integration of AR and gamified learning features

Summary

This methodology provides a structured framework for developing and evaluating an AR-based Solar System learning application. By combining immersive AR visualization, gamified learning activities, and a statistically validated evaluation method, the study effectively measures the impact of immersive technologies in STEM education.

2.2 Commercialization Aspect of the Product

The proposed Augmented Reality (AR)-based Solar System learning application is designed not only as a research prototype but also as a scalable and commercially viable educational technology solution. The system targets secondary education institutions in Sri Lanka, particularly schools delivering Grade 8 STEM curricula, as well as supplementary learning environments such as tuition classes and educational resource centers.

The commercialization strategy focuses on delivering a **low-cost, curriculum-aligned, and offline-capable mobile learning platform** that can be deployed without requiring expensive infrastructure or specialized hardware. Unlike many existing AR educational solutions that depend on high-end devices or continuous internet connectivity, this system is optimized for **low-end Android smartphones (2–3 GB RAM)** and functions fully in offline mode after initial installation. This makes it particularly suitable for resource-constrained educational environments across Sri Lanka.

Value Proposition

The system provides distinct value to key stakeholders:

For Schools and Educational Institutions

- Enhances STEM learning through interactive AR visualization of Solar System concepts
- Reduces dependency on physical laboratory equipment by providing virtual simulations
- Supports teachers with structured lesson modules and guided learning activities
- Enables digital transformation of science education with minimal infrastructure cost

For Students

- Improves conceptual understanding through immersive 3D visualization
- Encourages active learning through interaction, exploration, and gamified activities
- Supports multiple learning styles including visual, kinaesthetic, and experiential learning
- Increases engagement and motivation compared to traditional textbook-based learning

For Education Authorities and Policymakers

- Provides a curriculum-aligned digital learning tool that supports national STEM education goals
- Enhances classroom teaching effectiveness through blended learning approaches
- Contributes to improved student performance in science-related subjects
- Supports long-term educational digitalization initiatives in Sri Lanka

Deployment Strategy

The system is designed for flexible deployment across multiple educational settings:

- Installation via Android APK or future Google Play Store release
- Direct deployment in school computer labs or teacher-managed devices
- Integration into classroom teaching as a supplementary AR learning tool
- Training sessions for teachers to ensure effective classroom adoption

Scalability and Expansion

The application is designed with scalability in mind and can be extended in future developments to:

- Cover additional STEM subjects such as Physics, Chemistry, and Biology
- Expand from Grade 8 to other secondary school grade levels
- Introduce advanced AR modules such as space exploration simulations
- Support multilingual content delivery for wider national accessibility

Sustainability Model

Sustainability is ensured through a combination of institutional adoption and low-maintenance digital distribution. The system is designed to require minimal operational cost after deployment, with periodic updates aligned with curriculum revisions. Future updates can be delivered digitally without requiring hardware upgrades.

Market Positioning

Unlike many global AR educational tools that are not aligned with local curricula or require high-end infrastructure, this system addresses a critical gap in the Sri Lankan education sector by providing a **localized, affordable, and curriculum-specific AR learning solution**. This positions the product as a practical alternative for integrating

immersive technology into mainstream STEM education in developing-country contexts.

Overall, the AR-based Solar System learning application presents a strong commercialization potential due to its low-cost architecture, offline functionality, curriculum alignment, and adaptability to existing educational environments. By bridging the gap between traditional teaching methods and modern immersive technologies, the system offers a sustainable and scalable solution for enhancing STEM education in Sri Lanka.

2.3 Testing and Implementation

The testing and implementation phase of the proposed Augmented Reality (AR)-based Solar System learning application was carried out systematically to evaluate its functionality, performance, usability, and educational effectiveness. The implementation involved integrating all system components, including the marker-based AR engine, interactive 3D Solar System models, gamified learning activities, and in-application quiz modules, into a unified mobile application developed using Unity and Vuforia.

Functional Testing

Functional testing was conducted to ensure that all core features of the system operated correctly and consistently. This phase focused on verifying the reliability of the AR-based interactions and learning modules.

The following aspects were tested:

- Accurate recognition of AR image targets using the device camera
- Proper rendering and display of 3D Solar System models upon marker detection
- Smooth interaction with 3D objects, including rotation, scaling, and zooming
- Correct execution of gamified learning activities such as:
 - Planet Size Comparison

- Planet Order Arrangement
- Fact Matching

This stage ensured that all functional requirements were met and that the system behaved as intended without logical errors or crashes.

Performance Testing

Performance testing was carried out to evaluate the efficiency and stability of the application on mid-range Android devices, which are commonly used in Sri Lankan school environments. The main objective was to ensure smooth operation under limited hardware conditions.

To optimize performance, the following techniques were applied:

- Reduction of polygon count in 3D models to improve rendering speed
- Compression of textures to reduce memory usage
- Optimization of real-time AR rendering processes
- Efficient asset management for smoother application execution

These optimizations ensured that the application maintained stable performance without lag, delays, or unexpected crashes during usage.

Usability Testing

Usability testing was conducted with Grade 8 students to evaluate the user experience and ease of interaction with the AR system. Students were required to engage with the application by:

- Scanning AR markers from textbook diagrams
- Exploring interactive 3D Solar System models
- Participating in gamified learning activities
- Completing in-app quizzes designed to assess understanding

Observations were made regarding navigation simplicity, clarity of instructions, and overall engagement levels. The results indicated that students were able to easily understand and interact with the system, and they showed increased interest and motivation compared to traditional learning methods.

Implementation Process

The system was implemented in a controlled educational environment using a structured two-phase learning approach:

1. **Traditional Learning Phase:** Students first completed a paper-based assessment based on conventional textbook learning methods. This was used to measure baseline understanding of Solar System concepts.
2. **AR-Based Learning Phase:** Students then interacted with the AR mobile application, where they explored 3D models, completed gamified activities, and answered in-app quiz questions.

This dual-phase implementation allowed for a direct comparison between traditional and AR-based learning outcomes using the same group of students.

All student responses, quiz scores, and interaction data were recorded digitally and exported into spreadsheet format for further statistical analysis and evaluation.

CHAPTER 4: RESULTS AND DISCUSSION

This chapter presents the results of the system implementation, the findings from the user evaluation study, and a critical discussion of the results in the context of the research objectives. Limitations of the current implementation are also addressed.

4.1 Results

The results of this study indicate a clear improvement in student learning performance after the introduction of the Augmented Reality (AR)-based Solar System learning application. The analysis was conducted by comparing student performance under two conditions: traditional textbook-based learning and AR-enhanced learning using the developed mobile application.

Based on the collected data:

- Students achieved **higher post-test scores** after using the AR application compared to their pre-test (traditional learning) scores.
- A significant improvement was observed in questions related to:
 - Planet order and arrangement
 - Relative size of planets
 - Solar System structure and spatial understanding

The paired sample t-test analysis confirmed that there is a **statistically significant difference** between traditional learning and AR-based learning results, indicating that the AR system had a positive impact on student academic performance.

In addition to test scores, engagement data collected from the application showed:

- High participation rates in AR lessons and activities
- Increased completion rates of gamified learning modules
- Frequent interaction with 3D Solar System models

These results suggest that students were more actively engaged when using the AR-based learning system compared to traditional methods.

4.2 Research Findings

The findings of this study highlight several important outcomes regarding the effectiveness of Augmented Reality (AR) in enhancing STEM education, particularly in teaching abstract astronomical concepts at the Grade 8 level.

Firstly, the implementation of marker-based AR significantly improved students' **conceptual understanding** of complex Solar System topics. Abstract concepts such as planetary scale differences, orbital structures, and relative distances were more easily understood when represented through interactive 3D visualizations. This confirms that AR helps bridge the gap between theoretical knowledge and spatial understanding, which is often difficult to achieve through traditional textbook-based instruction.

Secondly, the study found a strong improvement in **student engagement and motivation**. The inclusion of gamified learning activities such as Planet Size Comparison, Planet Order Arrangement, and Fact Matching encouraged active participation rather than passive learning. Students showed greater interest in repeated interaction with the content, which contributed to deeper learning and improved knowledge retention.

Another key finding is the effectiveness of combining **AR visualization with interactive assessment methods**. The integration of in-application quizzes enabled immediate evaluation of student understanding, while interaction tracking provided additional insights into learning behavior. This continuous and performance-based assessment approach was found to be more effective than traditional examination-only methods, as it allowed real-time monitoring of student progress.

The study also revealed that students **preferred AR-based learning over traditional textbook-based methods**. The immersive and interactive nature of the system created a more engaging learning environment, resulting in sustained attention and higher interest in astronomy-related topics.

In addition, the **statistical analysis using a paired sample t-test** confirmed a significant improvement in post-test scores compared to pre-test scores. This provides strong quantitative evidence that the AR-based learning intervention had a positive impact on student academic performance.

Summary of Key Findings

- AR visualization significantly improved **conceptual understanding** of Solar System concepts.
- Gamified activities increased **student engagement and motivation**.
- Interactive AR learning improved **knowledge retention** compared to traditional methods.
- AR-based learning was found to be more **effective than textbook-based teaching** in STEM education.
- Statistical results (paired t-test) confirmed a **significant improvement in learning outcomes**.

4.3 Discussion

The findings of this study demonstrate that Augmented Reality (AR) has a significant positive impact on students' understanding and engagement in learning abstract astronomical concepts within the Grade 8 STEM curriculum. The improvement in student performance observed through the paired sample t-test supports the effectiveness of immersive learning compared to traditional textbook-based instruction.

One of the key reasons for this improvement is the ability of AR to transform abstract and spatially complex concepts into interactive 3D visual experiences. Topics such as planetary scale, orbital positioning, and relative distances are typically difficult for students to fully understand using static diagrams. However, the AR system allows learners to manipulate and explore Solar System models in real time, which enhances spatial cognition and supports the development of more accurate mental models. This

aligns with prior research which suggests that visualization-based learning significantly improves conceptual understanding in science education.

In addition, the integration of gamified learning elements contributed strongly to increased student motivation and engagement. Activities such as Planet Size Comparison, Planet Order Arrangement, and Fact Matching encouraged active participation and repeated interaction with the learning content. This shift from passive learning to active learning played an important role in improving knowledge retention and sustaining student interest throughout the learning process. The findings support existing literature that highlights gamification as an effective strategy for increasing learner motivation in digital learning environments.

Furthermore, the combination of AR visualization with in-application quizzes provided a more dynamic and continuous assessment approach. Unlike traditional evaluation methods that rely on final examinations, the system enabled real-time performance tracking and immediate feedback. This allowed students to identify and correct misunderstandings during the learning process itself, leading to better learning outcomes. The use of a paired sample t-test further validated that the improvement in performance was statistically significant, confirming the effectiveness of the AR-based intervention.

However, it is important to acknowledge certain limitations in this study. The evaluation was conducted within a limited sample size and controlled classroom environment, which may not fully represent diverse learning conditions across different schools. Additionally, long-term retention of knowledge was not assessed, which could be considered in future research to further validate the effectiveness of AR-based learning.

Despite these limitations, the results clearly indicate that Augmented Reality provides a more engaging and effective learning experience compared to traditional methods. The system demonstrates strong potential as a supplementary educational tool that can enhance STEM education by improving visualization, engagement, and conceptual understanding in a practical and scalable manner.

CHAPTER 5: CONCLUSION

This study investigated the effectiveness of Augmented Reality (AR) in enhancing STEM education, specifically in teaching Grade 8 Solar System concepts within the Sri Lankan school curriculum. The primary objective was to compare traditional textbook-based learning with an AR-based mobile learning application and evaluate its impact on student conceptual understanding, engagement, and learning outcomes using a paired sample t-test.

The findings of the study clearly demonstrate that the AR-based learning approach significantly improves student performance compared to traditional teaching methods. Students who interacted with the AR application showed higher post-test scores, indicating improved conceptual understanding of abstract astronomical topics such as planetary structure, orbital relationships, and relative scale. The statistical analysis further confirmed that this improvement was significant, validating the effectiveness of immersive learning technology in educational environments.

In addition to academic performance, the study also found that AR-based learning greatly enhances student engagement and motivation. The integration of interactive 3D visualizations and gamified learning activities encouraged active participation, sustained attention, and increased interest in learning. This shift from passive to active learning contributed to better knowledge retention and deeper understanding of the subject matter.

Furthermore, the study highlights that Augmented Reality provides a practical and effective solution for addressing limitations in traditional education, particularly the inability to visually represent dynamic and spatial scientific concepts using static textbook materials. The system developed in this research demonstrates that immersive technologies can successfully bridge this gap while remaining compatible with low-resource educational environments through offline functionality and optimization for low-end Android devices.

However, the study was conducted within a limited sample size and controlled environment, which may restrict the generalization of results across all educational settings. Future research can expand this work by involving a larger number of schools, different grade levels, and additional STEM subjects to further validate and enhance the system.

Overall, this research concludes that Augmented Reality is a highly effective educational tool that can significantly improve STEM learning outcomes. The developed AR-based mobile application provides a scalable, engaging, and curriculum-aligned solution that has the potential to transform traditional teaching methods into more interactive and student-centered learning experiences in Sri Lanka.

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APPENDIX A: SYSTEM MODULE DESCRIPTIONS

This appendix provides a detailed description of the main modules implemented in the Augmented Reality (AR)-based Solar System learning application. Each module has been designed to support interactive learning, improve student engagement, and enable effective assessment of conceptual understanding in STEM education.

A.1 Home Module

The Home Module acts as the main entry point of the application. It provides navigation access to all major sections, including lessons, AR exploration, gamified activities, and quizzes. The interface is designed with simplicity and clarity to ensure ease of use for Grade 8 students, enabling smooth interaction with the system.

A.2 AR Exploration Module

This module represents the core functionality of the application. It implements a marker-based Augmented Reality system where predefined image targets from textbooks are scanned using a mobile device camera. Once detected, corresponding 3D models of the Solar System, including the Sun and planets, are rendered in real time.

Users can:

- Rotate 3D planetary models
- Zoom in and out for detailed observation
- Interact with celestial objects in a 3D space

This module significantly enhances spatial understanding by transforming static textbook diagrams into immersive and interactive visual learning experiences.

A.3 Lesson Module

The Lesson Module provides structured educational content aligned with the Grade 8 STEM curriculum. The content is organized into three main sections:

- Solar System & Sun
- Inner Planets
- Outer Planets

Each section includes clear explanations supported by AR visualizations, allowing students to connect theoretical knowledge with interactive 3D representations. This improves conceptual clarity and knowledge retention.

A.4 Gamified Learning Module

The Gamified Learning Module is designed to enhance student engagement through interactive learning activities. It includes educational mini-games such as:

- **Planet Size Comparison:** Students compare relative sizes of planets
- **Planet Order Arrangement:** Students arrange planets in correct orbital order
- **Fact Matcher:** Students match scientific facts with corresponding planets

These activities encourage active participation, reinforce learning outcomes, and promote long-term knowledge retention through game-based learning strategies.

A.5 Quiz and Assessment Module

The Quiz Module is used to evaluate student understanding after completing lessons and gamified activities. It consists of structured questions related to Solar System concepts, including multiple-choice and interactive question formats.

Student responses are automatically recorded within the system, allowing for accurate assessment of learning outcomes and supporting further statistical analysis of performance improvements.

A.6 Data Logging and Performance Tracking Module

This module is responsible for capturing and storing user interaction data during application usage. The collected data includes:

- Quiz scores and assessment results
- Game performance metrics
- Interaction frequency with AR models
- Task completion time

This data is exported into spreadsheet format for further analysis, enabling the evaluation of student engagement patterns and system effectiveness in improving learning outcomes.

A.7 Settings Module

The Settings Module allows users to customize application preferences to improve usability. It includes options such as sound control, navigation settings, and general interface preferences. This ensures a more flexible and user-friendly learning experience for students.

APPENDIX B: USER EVALUATION QUESTIONNAIRE

APPENDIX C: SAMPLE AI-GENERATED SHOTLIST JSON